8

The influence of the non-greyness of the absorption coefficient

8.1 The continuum energy distribution

We have seen earlier that for a linear source function

$$F_\lambda = S_\lambda(\tau_\lambda = \tfrac{2}{3}), \qquad 8.1$$

which is called the Eddington–Barbier relation. For the grey case when $\tau_\lambda = \tau$, the layer with $\tau_\lambda = \frac{2}{3}$ always refers to the same geometrical depth. But if κ_λ is wavelength dependent this is no longer the case. Look at the κ_λ in a B0 star (Fig. 7.9). At $\lambda = 3647^+$ we have a lower than average κ_λ; at $\lambda = 3647^-$ the κ_λ is much higher than average. In Fig. 8.1 we demonstrate the situation.

At $\lambda = 3647^+$, i.e., on the long wavelength side of the discontinuity, the layer with $\tau_\lambda = \frac{2}{3}$ refers to a geometrical depth t_λ which is

$$t_\lambda = \frac{2}{3}\frac{1}{\kappa_\lambda(3647^+)}, \qquad 8.2$$

and which is larger than the average geometrical depth seen, namely

$$t = \frac{2}{3}\frac{1}{\bar{\kappa}}, \qquad 8.3$$

where $\bar{\kappa}$ is the average absorption coefficient, averaged over wavelengths.

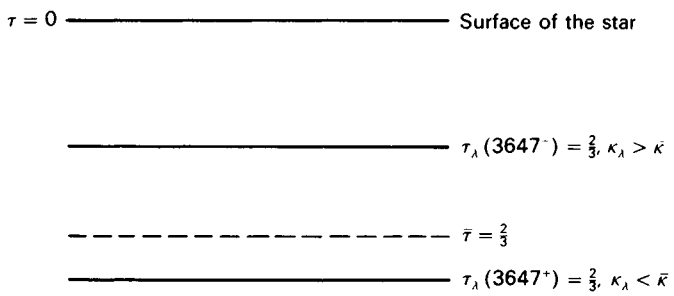

Fig. 8.1. For wavelengths with large κ_λ we see high layers with low temperatures. For wavelengths with small κ_λ we see deep layers with high T. $\bar{\tau}$ is the optical depth corresponding to $\bar{\kappa}$.

Thus, at this wavelength of 3647^+ Å, we obtain radiation from a deeper than average geometrical depth. At the deeper layer the source function and the temperature are larger and so we get more radiation at this wavelength with a small κ_λ than we would have received in a grey atmosphere with $\kappa = \bar{\kappa}$.

On the other hand, at $\lambda = 3647^-$ Å, i.e., at the short wavelength side of the Balmer discontinuity, where κ_λ is larger than $\bar{\kappa}$, we receive radiation from a higher level, where T and S_λ are smaller than at $\bar{\tau} = \frac{2}{3}$, and we receive less radiation than for a grey atmosphere with $\kappa = \bar{\kappa}$. Here $\bar{\tau}$ is the optical depth corresponding to $\bar{\kappa}$.

We show the relation between the wavelength dependence of κ_λ and F_λ in Fig. 8.2. The distribution of the flux F_λ looks like a mirror image of the wavelength distribution of κ_λ. We find discontinuities in the continuum energy distribution at those wavelengths where κ_λ has a discontinuity.

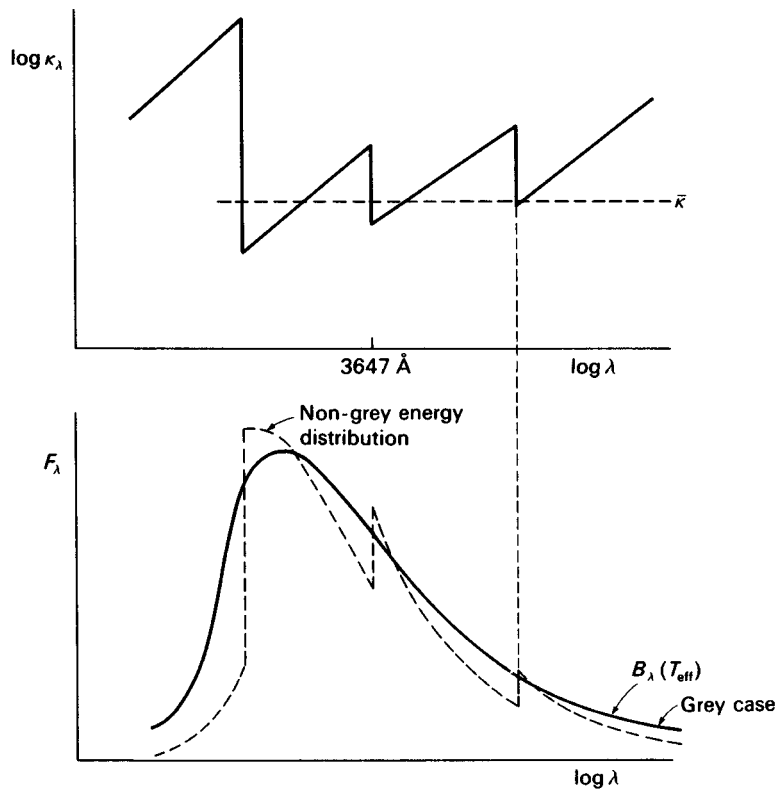

Fig. 8.2. Schematic diagram showing the influence of the non-greyness of the hydrogen absorption coefficient on the observed energy distribution of the star.

8.2 The dependence of the Balmer discontinuity on temperature and electron density

For F and G stars the discontinuity at the Balmer absorption edge is determined by the ratio of the absorption coefficient at the long wavelength side, which is due to the H^- absorption, to the absorption coefficient on the short wavelength side, which is due to the sum of the H^- absorption and the H absorption from the second quantum level (see Fig. 8.3). This means

$$\frac{\kappa(3647^+)}{\kappa(3647^-)} = \frac{\kappa(H^-)N(H^-)}{\kappa(H^-)N(H^-) + \kappa^-(H)N_H(n=2)}. \qquad 8.4$$

Here $\kappa^-(H)$ and $\kappa(H^-)$ are the hydrogen and H^- absorption coefficients per H atom in the level $n = 2$ and per H^- ion respectively at the short wavelength side of 3647 Å. From the Saha equation we know that

$$N(H^-) = f(T)N_H(n=1)n_e,$$

where n_e is the number of free electrons per cm^3, also called the *electron density*, and $f(T)$ is a slowly varying function of T. We find then that

$$\frac{\kappa(3647^+)}{\kappa(3647^-)} = \frac{\kappa(H^-)N_H(n=1)n_e f(T)}{\kappa(H^-)N_H(n=1)n_e f(T) + \kappa^-(H)N_H(n=2)}. \qquad 8.5$$

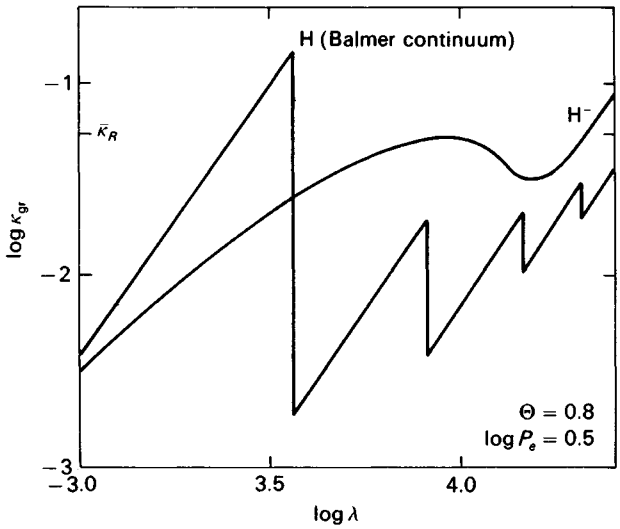

Fig. 8.3. The continuous absorption coefficients κ per gram of hydrogen and the negative hydrogen ion H^- for a temperature of 6300 K and an electron pressure of $\log P_e = 0.5$. For $\lambda < 3647$ Å, κ is mainly determined by neutral hydrogen, while for $\lambda > 3647$ Å, κ is determined by H^-.

Generally, the discontinuity is a measure of the temperature and the electron density, and can be used to determine one of these quantities.

For F stars the contribution of H^- to the absorption on the short wavelength side of $\lambda = 3647$ Å is very small. In this case, equation (8.5) reduces to

$$\frac{\kappa(3647^+)}{\kappa(3647^-)} = \frac{\kappa(H^-)N_H(n=1)n_e f(T)}{\kappa^-(H)N_H(n=2)} \propto \frac{N_H(n=1)}{N_H(n=2)} n_e. \qquad 8.6$$

For cool stars the Balmer discontinuity depends on both the temperature and the electron density. If we can determine the temperature from another observation we can determine the electron density from this discontinuity. With increasing electron density the discontinuity becomes smaller because the H^- absorption increases when more H^- ions are formed for higher electron densities. In Fig. 8.4 we compare the energy distributions of an

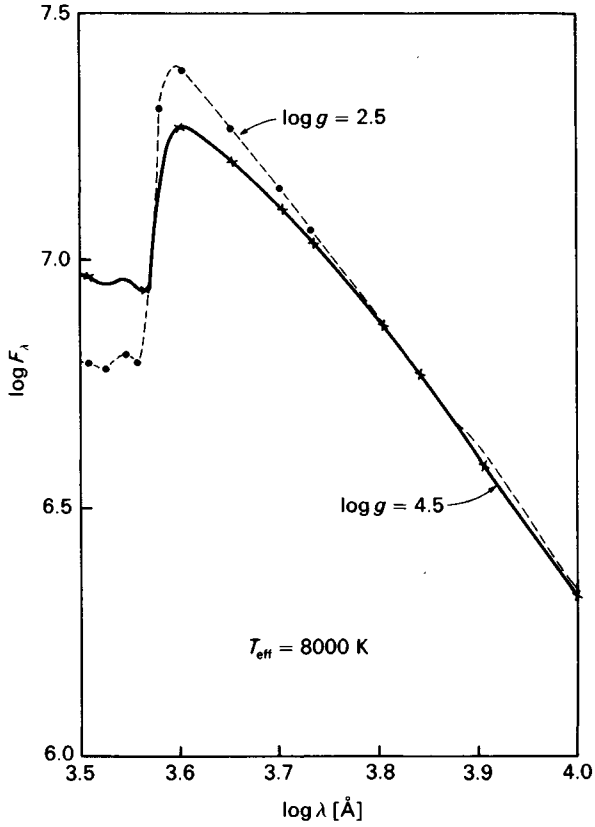

Fig. 8.4. The radiative flux F_λ [erg cm^{-2} s^{-1} Å^{-1}] is shown as a function of wavelength for a main sequence star with $\log g = 4.5$ (solid line) and for a bright giant with $\log g = 2.5$ (dashed curve). The bright giant shows a much larger Balmer jump because of the smaller contribution of H^- to the absorption coefficient.

F0 V star with one for an F0 II bright giant. The Balmer discontinuity is larger for the bright giant.

For stars hotter than 9000 K the H^- absorption is negligible in comparison with the hydrogen absorption. The absorption coefficient on the long wavelength side of the Balmer discontinuity, also called the *Balmer jump*, is due to the absorption from hydrogen atoms in the third quantum level, while that on the short wavelength side is mainly due to hydrogen absorption from atoms in the second quantum level. The ratio of the absorption coefficients then becomes

$$\frac{\kappa(3647^+)}{\kappa(3647^-)} \approx \frac{\kappa^+(\mathrm{H})N_{\mathrm{H}}(n=3)}{\kappa^-(\mathrm{H})N_{\mathrm{H}}(n=2)}, \qquad 8.7$$

where $\kappa^+(\mathrm{H})$ is the hydrogen absorption coefficient per absorbing atom at the long wavelength side of the Balmer jump and $\kappa^-(\mathrm{H})$ that on the short wavelength side. For the higher temperatures the Balmer jump depends only on the ratio of the hydrogen atoms in the second and in the third quantum levels, which, according to the Boltzmann formula, is only a function of temperature. *For hot stars the temperature can therefore be determined from the Balmer jump alone.*

For yet higher temperatures and for supergiants the electron scattering becomes important first on the long wavelength side and for still higher temperatures also on the short wavelength side of the Balmer discontinuity. This reduces the Balmer discontinuity. In this case the discontinuity depends again on both the temperature and the electron density.

8.3 The influence of the Balmer jump on the UBV colors

Fig. 1.7 shows the two-color diagram for black bodies of different temperatures. For a grey atmosphere for which we obtain radiation from the same depth $t(\tau = \frac{2}{3})$ at all wavelengths we would see radiation from the temperature at this depth at all wavelengths. If $S_\lambda = B_\lambda(T)$, then the observed energy distribution should be that of a black body with the temperature of that layer. As we saw in Section 6.4, the temperature in $\tau = \frac{2}{3}$ is equal to the effective temperature. For a grey atmosphere the energy distribution should therefore correspond to that of a black body with $T = T_{\mathrm{eff}}$. A look at Fig. 1.7 shows that the colors of the stars do not correspond to those of black bodies. Obviously, the stars do not have grey atmospheres. We have just seen how large the variations of the continuous absorption coefficients can be. In Fig. 8.2 we see how the actual stellar energy distributions deviate from those of black bodies. We receive more radiation for the wavelength regions where κ_λ is small, which is generally

the case on the long wavelength side of the Balmer discontinuity. In the blue region therefore, we generally receive more energy than we would in the case of a grey atmosphere. This is not the case in the visual region. We therefore expect non-grey atmospheres to look more blue than the grey ones, at least if we consider only the energy distribution in the continuum.

For wavelengths shorter than 3647 Å the energy is reduced by large factors when we look at the spectra of B, A, and F stars. This is due to the strong absorption from the hydrogen atoms in the second quantum level for these stars. The U filter, used to measure the U colors, is sensitive for radiation on both sides of the Balmer discontinuity. While there is somewhat more radiation on the long wavelength side of the Balmer discontinuity, the flux on the short wavelength side is greatly reduced. This reduction by far outweighs the increase on the long wavelength side. Because of the Balmer discontinuity the U – B colors become redder, i.e., more positive, while the B – V colors become smaller.

The changes of the colors are largest for A stars with the strong Balmer jump. In Fig. 8.5 we have qualitatively indicated by solid arrows which way the continuum colors change due to the Balmer absorption.

Additional strong variations of κ_λ with wavelength are introduced by the spectral lines in which the absorption coefficient also changes by very large factors. While we will discuss the formation of spectral lines in Chapters 10 and 11, we discuss here which color changes are introduced due to the spectral lines. We have already seen that they are strongest for the reddest or coolest stars. The colors of these stars are therefore strongly influenced by the spectral line absorption. We also observe that there are more lines in the blue region than in the visual and that there are still more lines in the ultraviolet. The line absorption therefore tends to make all colors more red. This color change is superimposed on the color change due to the variation in the continuous κ. It becomes very strong for stars with spectral types later than F. It is of little influence for stars with spectral types earlier than F. In Fig. 8.5 we have indicated by dashed arrows the color changes for cool stars due to line absorption.

8.4 The influence of the non-greyness on the temperature stratification

8.4.1 The mean absorption coefficient $\bar{\kappa}$

We may expect that the grey temperature stratification is still a good approximation if we choose an appropriate mean value for the $\bar{\kappa}$ and the corresponding $\bar{\tau}$. How do we have to determine $\bar{\kappa}$ in order to make

the grey temperature stratification a good approximation? Since the flux determines the temperature stratification, the flux ought to come out correctly. So we must require that $\int_0^\infty F_\lambda \, d\lambda$ is the same for the non-grey case and in the grey approximation.

As in the grey case, we want to write

$$S = B = \tfrac{3}{4}F(\bar{\tau} + q(\bar{\tau})) \simeq \tfrac{3}{4}F(\bar{\tau} + \tfrac{2}{3}) \qquad \text{or} \qquad F = B(\bar{\tau} = \tfrac{2}{3}). \tag{8.8}$$

For a linear source function, i.e., $S_\lambda = a + b\tau_\lambda$, we had seen earlier that at the surface

$$F = \int_0^\infty F_\lambda \, d\lambda = \int_0^\infty B_\lambda(\tau_\lambda = \tfrac{2}{3}) \, d\lambda. \tag{8.9}$$

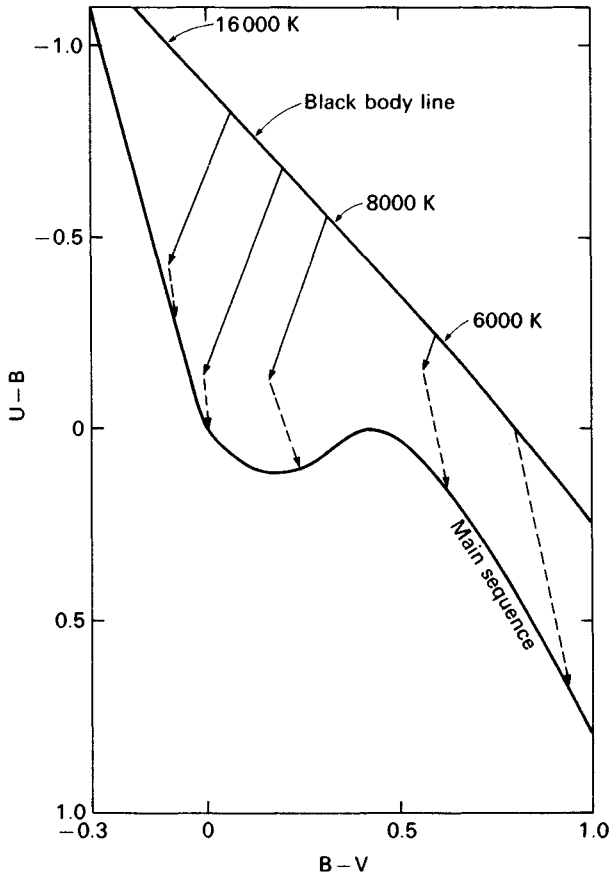

Fig. 8.5. The two-color diagram is shown for main sequence stars and for black bodies. The solid arrows show qualitatively how the colors of grey stars (black bodies) change due to the Balmer discontinuity. The dashed arrows indicate how the line absorption changes the colors qualitatively. For hot stars the Balmer discontinuity is more important than the line absorption. For cool stars the line absorption becomes very important, while the Balmer discontinuity disappears.

This should then be equal to what is obtained in the grey case:

$$\int_0^\infty F_\lambda \, \mathrm{d}\lambda = \int_0^\infty B_\lambda(\bar{\tau} = \tfrac{2}{3}) \, \mathrm{d}\lambda. \qquad 8.10$$

As can be seen from Fig. 8.6, we can describe $B_\lambda(\tau_\lambda = \frac{2}{3})$ by a Taylor expansion around the layer $\bar{\tau} = \frac{2}{3}$. One then obtains

$$B_\lambda(\tau_\lambda = \tfrac{2}{3}) = B_\lambda(\bar{\tau} = \tfrac{2}{3}) + \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}} \Delta\bar{\tau}, \qquad 8.11$$

where the derivative has to be taken at the depth $\bar{\tau} = \frac{2}{3}$. Assuming $\kappa_\lambda/\bar{\kappa}$ to be the depth independent, we derive

$$\frac{\tau_\lambda}{\bar{\tau}} = \frac{\kappa_\lambda}{\bar{\kappa}} \qquad \text{and} \qquad \bar{\tau} = \tau_\lambda \frac{\bar{\kappa}}{\kappa_\lambda}. \qquad 8.12$$

We then find that at $\tau_\lambda = \frac{2}{3}$

$$\bar{\tau} = \frac{2}{3} \frac{\bar{\kappa}}{\kappa_\lambda} = \frac{2}{3} + \Delta\bar{\tau}, \qquad 8.13$$

according to the definition of $\Delta\bar{\tau}$ (see Fig. 8.6). From this equation we derive

$$\Delta\bar{\tau} = \frac{2}{3} \frac{\bar{\kappa}}{\kappa_\lambda} - \frac{2}{3} = \frac{2}{3}\left(\frac{\bar{\kappa}}{\kappa_\lambda} - 1\right). \qquad 8.14$$

The condition (8.9) for F_λ then requires

$$\begin{aligned} \int_0^\infty B_\lambda(\tau_\lambda = \tfrac{2}{3}) \, \mathrm{d}\lambda &= \int_0^\infty B_\lambda(\bar{\tau} = \tfrac{2}{3}) \, \mathrm{d}\lambda + \int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}} \times \frac{2}{3}\left(\frac{\bar{\kappa}}{\kappa_\lambda} - 1\right) \mathrm{d}\lambda \\ &= \int_0^\infty B_\lambda(\bar{\tau} = \tfrac{2}{3}) \, \mathrm{d}\lambda. \end{aligned} \qquad 8.15$$

For this to hold, we must require

$$\int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}} \frac{2}{3}\left(\frac{\bar{\kappa}}{\kappa_\lambda} - 1\right) \mathrm{d}\lambda = 0 \qquad \text{or} \qquad \int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}} \frac{\bar{\kappa}}{\kappa_\lambda} \mathrm{d}\lambda = \int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}} \mathrm{d}\lambda, \qquad 8.16$$

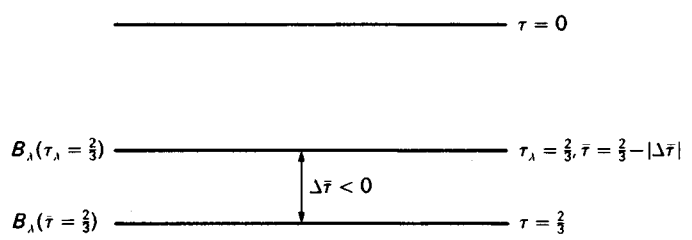

Fig. 8.6 demonstrates that $B_\lambda(\tau_\lambda = \frac{2}{3}) = B_\lambda(\bar{\tau} = \frac{2}{3}) + \frac{\mathrm{d}B_\lambda}{\mathrm{d}\tau_\lambda} \Delta\bar{\tau}(\tau_\lambda = \frac{2}{3})$, with $\Delta\bar{\tau} < 0$.

or, after dividing by $\bar{\kappa}$,

$$\frac{1}{\bar{\kappa}}\int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}}\,\mathrm{d}\lambda = \int_0^\infty \frac{1}{\kappa_\lambda}\frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}}\,\mathrm{d}\lambda \qquad \text{with } \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}} = \frac{\mathrm{d}B_\lambda}{\mathrm{d}T}\frac{\mathrm{d}T}{\mathrm{d}\bar{\tau}}. \tag{8.17}$$

This is the prescription for the determination of $\bar{\kappa}$, which we have derived here for the surface, for a linear depth dependence of the source function, and depth independent $\kappa_\lambda/\bar{\kappa}$.

The same condition can be derived for deeper layers if we start from the transfer equation (5.3) after integration over $\mathrm{d}\omega$:

$$\frac{\mathrm{d}}{\mathrm{d}\tau_\lambda}\tfrac{1}{4}F_\lambda = J_\lambda - S_\lambda; \tag{8.18}$$

or, if we multiply first by $\cos\vartheta$ and then integrate over ω and divide by π we find with the Eddington approximation $K_\lambda = \frac{1}{3}J_\lambda$ (which holds for an isotropic radiation field) that

$$\frac{\mathrm{d}K_\lambda}{\mathrm{d}\tau_\lambda} = \tfrac{1}{4}F_\lambda = \frac{1}{3}\frac{\mathrm{d}B_\lambda}{\mathrm{d}\tau_\lambda}, \tag{8.19}$$

for a deeper layer, where $J_\lambda \approx B_\lambda$. Or, rewriting the second half of the equation, we find

$$F_\lambda = \frac{4}{3}\frac{\mathrm{d}B_\lambda}{\mathrm{d}\tau_\lambda}. \tag{8.20}$$

This is a very important relation between F_λ and the gradient of the source function.

From the condition that

$$\int_0^\infty F_\lambda\,\mathrm{d}\lambda = F \text{ (grey)}, \tag{8.21}$$

we now derive

$$\frac{3}{4}\int_0^\infty F_\lambda\,\mathrm{d}\lambda = \int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}\tau_\lambda}\,\mathrm{d}\lambda = \int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}\bar{\tau}}\,\mathrm{d}\lambda \qquad \text{with } \tfrac{3}{4}F = \frac{\mathrm{d}B}{\mathrm{d}\bar{\tau}}. \tag{8.22}$$

This yields

$$\int_0^\infty \frac{1}{\kappa_\lambda}\frac{\mathrm{d}B_\lambda}{\mathrm{d}t}\,\mathrm{d}\lambda = \frac{1}{\bar{\kappa}}\int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}t}\,\mathrm{d}\lambda, \tag{8.23}$$

which is the same prescription as above. Using again

$$\frac{\mathrm{d}B_\lambda}{\mathrm{d}t} = \frac{\mathrm{d}B_\lambda}{\mathrm{d}T}\frac{\mathrm{d}T}{\mathrm{d}t} \qquad \text{and} \qquad \frac{\mathrm{d}B}{\mathrm{d}t} = \frac{\mathrm{d}B}{\mathrm{d}T}\frac{\mathrm{d}T}{\mathrm{d}t},$$

we derive again

$$\frac{1}{\bar{\kappa}} = \frac{\int_0^\infty \frac{1}{\kappa_\lambda}\frac{\mathrm{d}B_\lambda}{\mathrm{d}T}\,\mathrm{d}\lambda}{\int_0^\infty \frac{\mathrm{d}B_\lambda}{\mathrm{d}T}\,\mathrm{d}\lambda} = \int_0^\infty \frac{1}{\kappa_\lambda} G'_\lambda(T)\,\mathrm{d}\lambda = \int_0^\infty \frac{1}{\kappa_\lambda} G(\alpha)\,\mathrm{d}\alpha, \qquad 8.24$$

where $G'_\lambda(T)$ is the Rosseland weighting function and $\alpha = h\nu/kT$. From this definition of $G(\alpha)$ we find

$$G(\alpha) = \frac{kT}{h}\frac{\lambda^2}{c}\frac{\mathrm{d}B_\lambda/\mathrm{d}T}{\mathrm{d}B/\mathrm{d}T} = \frac{15}{4\pi^4}\frac{\alpha^4 \mathrm{e}^\alpha}{(\mathrm{e}^\alpha - 1)^2}. \qquad 8.25$$

This kind of mean absorption coefficient is called the *Rosseland mean absorption coefficient*. It is a harmonic mean. Frequencies with small κ_λ are weighted most heavily, because the *Rosseland mean $\bar{\kappa}$ is a flux weighted mean* and the flux is largest in the regions of small κ_λ, as we see from equation (8.20) if we write it in the form

$$F_\lambda = \frac{1}{\kappa_\lambda}\frac{\mathrm{d}B_\lambda}{\mathrm{d}t}\frac{4}{3}. \qquad 8.26$$

Because of this, the Rosseland mean $\bar{\kappa}$ is always close to the smallest values of κ_λ. See Figs. 7.5 and 8.3, where we have indicated the Rosseland mean values $\bar{\kappa}_\mathrm{R}$.

With the prescription (8.24) we can calculate mean values of $\bar{\kappa}$ for all desired temperatures and electron pressures.

For a given temperature and electron pressure we can calculate the degree of ionization for all the atoms and ions present in stellar atmospheres. With this we can add up all the numbers of particles to obtain the total number of particles per cm^3 from which the gas pressure can be obtained. We can thus also determine $\bar{\kappa}$ as a function of temperature and gas pressure. In Fig. 8.7 we reproduce a plot showing $\bar{\kappa}$ as a function of the gas pressure for different values of the temperature. The values of $\Theta = 5040/T$ are given on each curve. For low temperatures the continuous κ is essentially due to H^- only. For a given temperature the average $\bar{\kappa}$ increases proportionally to the electron pressure. For very low pressures the electron scattering becomes important and prevents a further decrease of $\bar{\kappa}$. At temperatures above 6000 K hydrogen absorption becomes important for very low gas pressures. At temperatures above 10 000 K hydrogen absorption is important for all pressures considered here and gives a large increase in $\bar{\kappa}$. At still higher temperatures hydrogen ionizes and $\bar{\kappa}$ does not increase any more, but decreases with increasing T. Finally, electron scattering takes over.

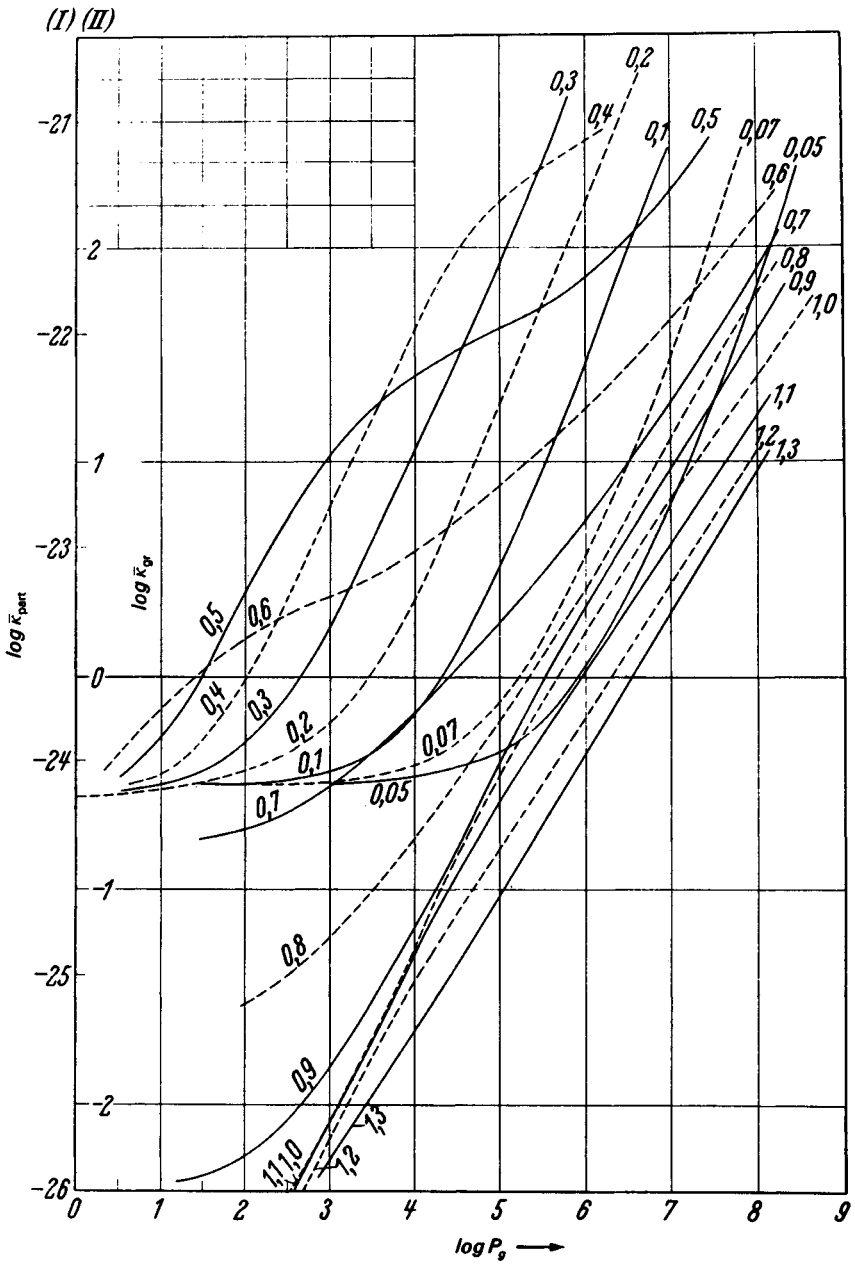

Fig. 8.7. The Rosseland mean values $\bar{\kappa}_R$ for the continuous absorption coefficient are shown as a function of the gas pressure. Each curve corresponds to a given value of the temperature. The value of $\Theta = 5040/T$ is given for each curve. The plot is double logarithmic. What is shown is actually the absorption coefficient per gram of material, i.e., for a column of gas which contains one gram of material, or per heavy particle, as shown on the outer left-hand scale. Solar element abundances were assumed.

8.4.2 *Changes in the temperature stratification $T(\bar{\tau})$ due to the non-greyness*

By computing $\bar{\kappa}$ in such a way as to make the integrated flux the same for the grey and for the non-grey case, we hoped that the temperature stratification

$$T^4 = \tfrac{3}{4}T_{\text{eff}}^4(\bar{\tau}_{\text{R}} + q(\bar{\tau}_{\text{R}})), \tag{8.27}$$

with $\bar{\tau}_{\text{R}}$ being the Rosseland mean value of τ_λ, i.e., with

$$\bar{\tau}_{\text{R}} = \int_0^t \bar{\kappa}_{\text{R}}\, \mathrm{d}t, \tag{8.28}$$

will still give a reasonable approximation. This is true qualitatively. Some important modifications, however, have to be taken into account.

We can see what happens if we look at the condition for radiative equilibrium

$$\int_0^\infty \kappa_\lambda J_\lambda\, \mathrm{d}\lambda = \int_0^\infty \kappa_\lambda S_\lambda\, \mathrm{d}\lambda \rightarrow = \int_0^\infty \kappa_\lambda B_\lambda\, \mathrm{d}\lambda \qquad \text{for LTE.} \tag{8.29}$$

If κ_λ varies strongly with λ, then these integrals are mainly determined by the wavelengths with *large* κ_λ. What happens to J_λ when κ_λ becomes very large as compared to the grey case? Let us look at the surface first, where $I_\lambda(\cos\vartheta) \sim B_\lambda(\tau_\lambda = \cos\vartheta)$. With the large κ_λ, the I_λ now comes from higher layers and therefore is smaller. So J_λ will also become smaller, but we must require $\int_0^\infty \kappa_\lambda J_\lambda\, \mathrm{d}\lambda = \int_0^\infty \kappa_\lambda B_\lambda\, \mathrm{d}\lambda$. If J_λ becomes smaller, B_λ must also become smaller at the surface, which means the temperature must decrease at the surface.

Therefore, generally the first effect of the non-greyness is a decrease in surface temperature; this is also called *surface cooling* and is due to the larger emission because of the larger κ_λ (though under special circumstances this may not be the case as for instance for strong lines in the red).

The second effect is the *backwarming*. We now have essentially many strong lines in the spectrum. In the regions with large κ_λ the intensity I_λ and the flux F_λ decrease, which means the total flux coming out of an atmosphere with the same temperature stratification is decreased. This means the same temperature stratification now leads to a smaller effective temperature. If we want to obtain the same total flux, we have to get more flux out of the wavelengths with the small κ_λ. If we have mainly many lines the flux has to be obtained from the wavelengths between the lines. This can only be done by increasing the temperature in the atmosphere.

Some people call this redistribution of energy: the energy absorbed at some wavelengths comes out at other wavelengths. But remember, this can

only be done by way of increasing the temperature in deeper layers. The energy distribution in the continuum now resembles the energy distribution of a somewhat hotter star, if the non-greyness is due mainly to line absorption. The star cannot absorb the energy at one wavelength and arbitrarily emit it at another wavelength, as we please.

A modified temperature distribution of the form

$$T^4 = \tfrac{3}{4}T'^4_{\text{eff}}(\bar{\tau} + q(\bar{\tau})) \qquad 8.30$$

leads to a flux

$$\int_0^\infty \pi F_\lambda \, \mathrm{d}\lambda = \pi F = \sigma T'^4_{\text{eff}}(1-\eta), \qquad 8.31$$

if η is the fraction of energy absorbed by the lines. If we require $\pi F = \sigma T^4_{\text{eff}}$ as before, we must have

$$T^4_{\text{eff}} = T'^4_{\text{eff}}(1-\eta) \qquad \text{or} \qquad T'^4_{\text{eff}} = \frac{T^4_{\text{eff}}}{1-\eta}. \qquad 8.32$$

In the case of the sun, $\eta \sim 14\%$, so $(1-\eta) = 0.86$ and

$$T'_{\text{eff}} = T_{\text{eff}}/\sqrt[4]{0.86} \approx T_{\text{eff}}(1+0.037). \qquad 8.33$$

In the sun with $\eta \sim 0.14$ the T'_{eff} must be higher than T_{eff} by about 3.5%, or by about 200 degrees. In the continuum the atmosphere looks like one

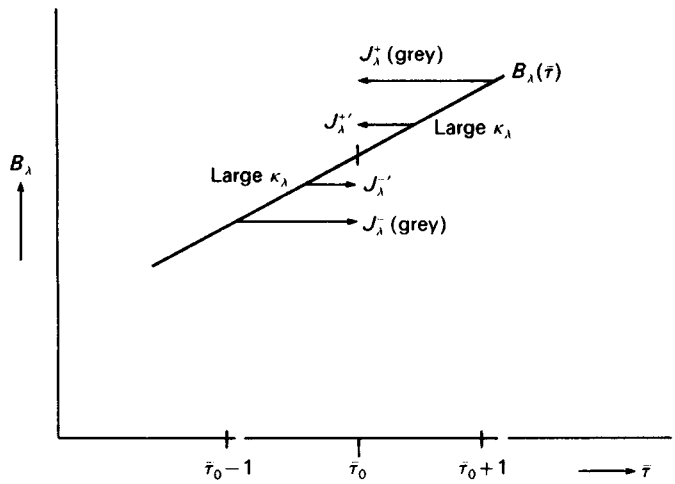

Fig. 8.8. In wavelengths with large κ_λ the ingoing average intensity J_λ^- is increased by ΔJ^- because the radiation originates at a deeper point with a larger Planck function than in the grey case. The outgoing radiation originates at a higher layer with a smaller Planck function than in the grey case. The average intensity J_λ^+ is reduced by the amount ΔJ^+ as compared to the grey case. If the Planck function is a linear function of depth $\Delta J_\lambda^+ = -\Delta J_\lambda^-$ and for deep layers the sum $J_\lambda^+ + J_\lambda^-$ remains the same as in the grey case.

with a temperature increased by about 200 degrees. With an effective temperature of 5800 K for the sun the energy distribution in the continuum looks similar to that of a black body of 6000 K, as we saw earlier in Volume 1.

What happens to the J_λ in the deep layers?

For wavelengths with small κ_λ the outgoing average intensity J_λ^+ is increased because it now originates in deeper layers than in the grey case, but the inward going intensity now originates at a higher layer with a smaller Planck function. The sum of the ingoing and outgoing intensities remains the same as long as we are far enough away from the surface such that the Planck function is still a linear function of depth. See Fig. 8.8 for the case of large κ_λ.